

HEARING LOSS (Classification Figure 1)

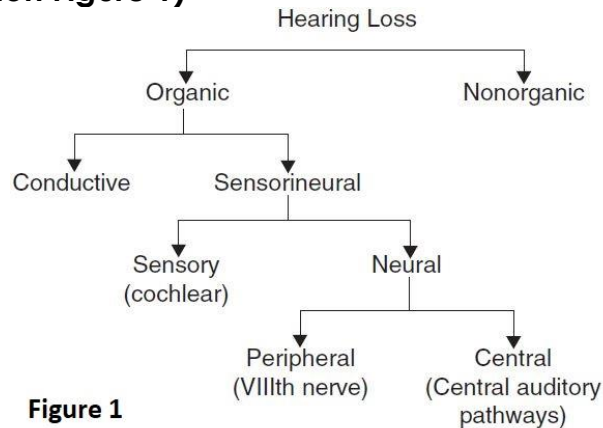


Figure 1

CONDUCTIVE HEARING LOSS AND ITS MANAGEMENT

Any disease process which interferes with the conduction of sound to reach the cochlea causes conductive hearing loss. The lesion may lie in the external ear and tympanic membrane, middle ear or ossicles up to the stapedio-vestibular joint.

The characteristics of conductive hearing loss are:

1. Negative Rinne test, i.e. BC > AC.
 2. Weber lateralised to the poorer ear.
 3. Normal absolute bone conduction.
 4. Low frequencies affected more.
 5. Audiometry shows bone conduction better than air conduction with an air-bone gap.
- The greater the air-bone gap, more is the conductive loss (Figure 2).
6. Loss is not more than 60 dB.
 7. Speech discrimination is good.

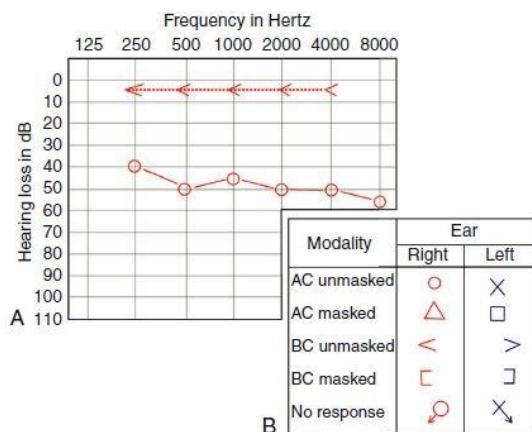


Figure 2 (A) Audiogram of right ear showing conductive hearing loss with A-B gap. (B) Symbols used in audiogram charting.

AETIOLOGY

The cause may be congenital (Table 1) or acquired (Table 2).

Table 1 Congenital causes of conductive hearing loss

- Meatal atresia
- Fixation of stapes footplate
- Fixation of malleus head
- Ossicular discontinuity
- Congenital cholesteatoma

Table 2 Acquired causes of conductive hearing loss

External ear	Any obstruction in the ear canal, e.g. wax, foreign body, furuncle, acute inflammatory swelling, benign or malignant tumour or atresia of canal.
Middle ear	(a) Perforation of tympanic membrane, traumatic or infective (b) Fluid in the middle ear, e.g. acute otitis media, serous otitis media or haemotympanum (c) Mass in middle ear, e.g. benign or malignant tumour (d) Disruption of ossicles, e.g. trauma to ossicular chain, chronic suppurative otitis media, cholesteatoma (e) Fixation of ossicles, e.g. otosclerosis, tympanosclerosis, adhesive otitis media (f) Eustachian tube blockage, e.g. retracted tympanic membrane, serous otitis media

AVERAGE HEARING LOSS SEEN IN DIFFERENT LESIONS OF THE CONDUCTIVE APPARATUS

1. Complete obstruction of the ear canal: 30 dB
2. Perforation of tympanic membrane (It varies and is directly proportional to the size of perforation): 10–40 dB
3. Ossicular interruption with intact drum: 54 dB
4. Ossicular interruption with perforation: 38 dB
5. Malleus fixation: 10–25 dB
6. Closure of oval window: 60 dB

Note here that ossicular interruption with an intact drum causes more loss than ossicular interruption with a perforated drum.

MANAGEMENT

Most cases of conductive hearing loss can be managed by medical or surgical means. Briefly, it consists of:

1. **Removal of canal obstructions**, e.g. impacted wax, foreign body, osteoma or exostosis, keratotic mass, benign or malignant tumors, or meatal atresia.
2. **Removal of fluid**. Myringotomy with or without grommet insertion.
3. **Removal of mass from middle ear**. Tympanotomy and removal of small middle ear tumors or cholesteatoma behind intact tympanic membrane.
4. **Stapedectomy**, as in osteosclerotic fixation of stapes footplate.
5. **Tympanoplasty**. Repair of perforation, ossicular chain or both.
6. **Hearing aid**. In cases where surgery is not possible, refused or has failed.

Otitis Media with Effusion (OME)

OME describes a non-purulent effusion in the middle ear space. The fluid varies from thin to mucoid. Mucoid fluid usually signifies a more chronic process. The fluid is secondary to obstruction of the eustachian tube.

Defining the cause of the obstruction is indicated prior to initiating therapy whenever possible. The varied etiologies include inflammation (bacterial, viral, allergic), congenital malformation, polyps or tumors of the nasopharynx, hypertrophied adenoids, cleft palate, radiation, endocrine, or iatrogenic. Serous fluid is commonly seen in the resolution stage of acute otitis media. OME is the most common cause of hearing loss in children. It is less frequent in adults and, when seen, the nasopharynx should be carefully evaluated for malignancy.

1. Signs and symptoms

The tympanic membrane may appear normal. Usually, it is slightly retracted even in early onset. Fluid, with or without bubbles, may be seen and maybe amber in color. With thicker fluid, the amber color is not prevalent, and the drum is dull. Prominence of the vessels is not infrequent, but in contrast to acute otitis media, the margins are distinct.

2. Diagnostic tests

Pneumatic otoscopy reveals diminished or no movement of the tympanic membrane. Tuning fork tests and audiometry usually reveal conductive hearing loss. Conductive loss on audiometry should not exceed 30-40 dB with serous otitis media.

3. Management

a) In older children autoinflation and politzerization of the eustachian tube are possible, sometimes affecting resolution. If the fluid results from a resolving otitis media, the process is usually self-limiting, 90% in 3 months. Decongestants and antihistamines have not proved to be effective management for OME. Recognizing that a small percentage of serous effusions contain bacteria or may be associated with nasopharyngeal or eustachian tube infection, often a 6- to 8-week therapeutic trial of an antibiotic (amoxicillin or TMP-SMZ) is warranted. Pressure-equalizing (PE) tubes (ventilation tubes) are indicated in refractory cases to alleviate hearing loss and arrest the development of permanent tympanic membrane and ossicle malfunction. Adenoidectomy is probably beneficial in selected instances of recurrent OME and acute otitis media.

b) For adults an underlying cause should be determined if possible. CT scans of the nasopharynx and sinuses are indicated to rule out infection or tumor. With a suggestive history, an allergy evaluation is indicated. Myringotomy with or without tube insertion may be indicated in selected instances.

SENSORINEURAL HEARING LOSS AND ITS MANAGEMENT

Sensorineural hearing loss (SNHL) results from lesions of the cochlea, 8th cranial nerve or central auditory pathways. It may be present at birth (congenital) or start later in life (acquired).

The characteristics of sensorineural hearing loss are:

1. A positive Rinne test, i.e. AC > BC.
2. Weber lateralised to a better ear.
3. Bone conduction reduced on Schwabach and absolute bone conduction tests.
4. More often it involves high frequencies.
5. No gap between air and bone conduction curve on audiometry (Figure 3).
6. Loss may exceed 60 dB.
7. Speech discrimination is poor.
8. There is difficulty in hearing in the presence of noise.

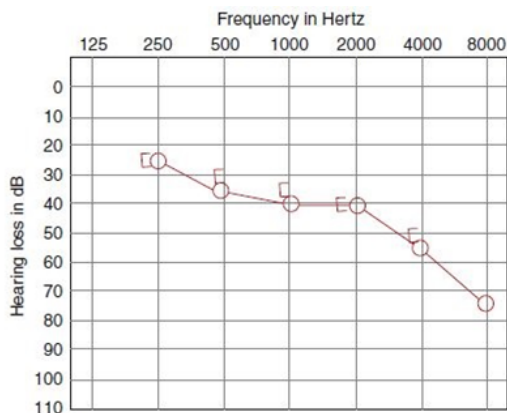


Figure 3 Audiogram of right ear showing sensorineural loss with no A-B gap.

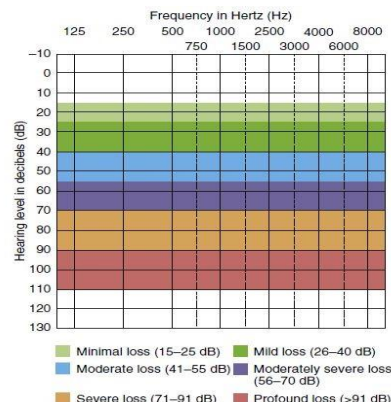


Figure 4 Classification of hearing loss. Ninety-five per cent of population has thresholds between 210 and 110 dB HL.

AETIOLOGY

CONGENITAL

It is present at birth and is the result of anomalies of the inner ear or damage to the hearing apparatus by prenatal or perinatal factors.

ACQUIRED

It appears later in life. The cause may be genetic or nongenetic. The genetic hearing loss may manifest late (delayed onset) and may affect only the hearing, or be a part of a larger syndrome affecting other systems of the body as well (syndromal).

Common causes of acquired SNHL include:

1. Infections of the labyrinth—viral, bacterial or spirochaetal
2. Trauma to labyrinth or 8th nerve, e.g. fractures of the temporal bone or concussion of the labyrinth or the ear surgery
3. Noise-induced hearing loss
4. Ototoxic drugs
5. Presbycusis
6. Ménière's disease
7. Acoustic neuroma
8. Sudden sensori-neural hearing loss
9. Familial progressive SNHL

10. Systemic disorders, e.g. diabetes, hypothyroidism, kidney disease, autoimmune disorders, multiple sclerosis, blood dyscrasias.

DIAGNOSIS

1. History. It is important to know whether the disease is congenital or acquired, stationary or progressive, associated with other syndromes or not, involvement of other members of the family and possible aetiological factors.

2. Severity of deafness (mild, moderate, moderately severe, severe, profound or total). This can be found in audiometry.

3. Type of audiogram. Whether loss is high frequency, low frequency, mid-frequency or flat type.

4. Site of the lesion, i.e. cochlear, retro-cochlear or central.

5. Laboratory tests. They depend on the etiology suspected, e.g. X-rays or CT scan of the temporal bone for evidence of bone destruction (congenital cholesteatoma, glomus tumor, middle ear malignancy, or acoustic neuroma), blood counts (leukemia), blood sugar (diabetes), serology for syphilis, thyroid functions (hypothyroidism), kidney function tests, etc.

MANAGEMENT

Early detection of SNHL is important as measures can be taken to stop its progress, reverse it, or to start an early rehabilitation program, which is essential for communication.

Syphilis of the inner ear is treatable with high doses of penicillin and steroids with improvement in hearing.

Hearing loss of *hypothyroidism* can be reversed with replacement therapy.

Serous labyrinthitis can be reversed by attention to middle ear infection.

Early management of *Ménière's disease* can prevent further episodes of vertigo and hearing loss.

SNHL due to *perilymph fistula* can be corrected surgically by sealing the fistula in the oval or round window with fat.

Ototoxic drugs should be used with care and discontinued if they cause hearing loss. In many such cases, it may be possible to regain hearing, total or partial, if the drug is stopped.

Noise-induced hearing loss can be prevented from further deterioration if the person is removed from the noisy surroundings.

DEGREE OF HEARING LOSS (WHO CLASSIFICATION)

WHO (1980) recommended the following classification based on pure tone audiogram taking the average of the thresholds of hearing for frequencies of 500, 1000, and 2000 Hz

DEGREE OF HEARING LOSS (Figure 4)

1. Mild 26–40 dB
2. Moderate 41–55 dB
3. Moderately severe 56–70 dB
4. Severe 71–91 dB
5. Profound More than 91 dB
6. Total

From this, it is implied that there is no *apparent* impairment of hearing from 0 to 25 dB.

SUDDEN SENSORI-NEURAL HEARING LOSS

Sudden SNHL is defined as 30 dB or more of SNHL over at least three contiguous frequencies occurring within a period of 3 days or less. Mostly it is unilateral. It may be accompanied by tinnitus or temporary spell of vertigo.

AETIOLOGY

Most often the cause of sudden deafness remains obscure, in which case it is called the idiopathic variety. In such cases, three aetiological factors are considered—viral, vascular or the rupture of cochlear membranes. Spontaneous perilymph fistulae may form in the oval or round window. Other aetiological factors which cause sudden deafness and must be excluded are listed below.

Remember the mnemonic “In The Very Ear Too No Major Pathology.”

1. Infections. Mumps, herpes zoster, meningitis, encephalitis, syphilis, otitis media.

2. Trauma. Head injury, ear operations, noise trauma, barotrauma, spontaneous rupture of cochlear membranes.

3. Vascular. Haemorrhage (leukaemia), embolism, or thrombosis of labyrinthine or cochlear artery or their vasospasm. They may be associated with diabetes, hypertension, polycythaemia, macroglobinaemia or sickle cell trait.

4. Ear (otologic). Ménière’s disease, Cogan’s syndrome, large vestibular aqueduct.

5. Toxic. Ototoxic drugs, insecticides.

6. Neoplastic. Acoustic neuroma. Metastases in cerebellopontine angle, carcinomatous neuropathy.

7. Miscellaneous. Multiple sclerosis, hypothyroidism, sarcoidosis.

8. Psychogenic.

MANAGEMENT

As far as possible, the aetiology of sudden hearing loss should be discovered by detailed history, physical examination and laboratory investigations. The investigations may include audiometry, vestibular tests, imaging studies of temporal bones, sedimentation rate, tests for syphilis, diabetes, hypothyroidism, blood disorders and lipid profiles. Some cases may require exploratory tympanotomy where perilymph fistula is strongly suspected. Where the cause still remains obscure, treatment is empirical and consists of:

1. Bed rest.

2. Steroid therapy. Prednisolone 40–60 mg in a single morning dose for 1 week and then tailed off in a period of 3 weeks. Steroids are anti-inflammatory and relieve oedema. They have been found useful in idiopathic sudden hearing loss of moderate degree.

3. Inhalation of carbogen (5% CO₂ + 95% O₂). It increases cochlear blood flow and improves oxygenation.

4. Vasodilator drugs.

5. Low molecular weight dextran. It decreases blood viscosity. It is contraindicated in cardiac failure and bleeding disorders.

6. Hyperbaric oxygen therapy. Available only in selected centres, hyperbaric oxygen raises the concentration of oxygen in labyrinthine fluids and improves cochlear function.

7. Low-salt diet and a diuretic. It is empirical and has same benefit as in cases of Ménière’s disease.

8. Intratympanic steroids therapy. It raises the local concentration of steroids in cochlear fluids, thus avoiding side effects of systemic therapy.

TREATMENT

Many treatment protocols have been suggested for idiopathic sensorineural sudden hearing loss but none has shown significant benefit over the benefit of spontaneous recovery which occurs in 50–60% cases within first 2 weeks.

None of the drugs, dextran 40, vasodilators, carbogen inhalation (5% CO₂ with 95% O₂), diatrizoate meglumine, have shown significant benefits.

Generally prescribed medicines include:

1. Steroids.
2. Inhalation of carbogen.
3. Low-salt diet and a diuretic.
4. Hyperbaric oxygen.

PROGNOSIS

Fortunately, about half the patients with idiopathic sensori-neural hearing loss recover spontaneously within 15 days. Chances of recovery are poor after 1 month. Severe hearing loss, and that associated with vertigo have poor prognosis. Younger patients below 40 and those with moderate losses have a better prognosis (see Table 3).

Table 3 Prognostic factors in sudden SNHL

Good prognosis	Bad prognosis
Mild loss	Severe loss
Low and medium frequency loss	High frequency loss
Recovery starting in 2 weeks	Recovery does not start in 2 weeks
No history of vertigo	History of vertigo
Younger patients	Older patients
Early treatment	Late treatment

